# KASBench Full Disclosure Report

## Preamble

This Full Disclosure Report (FDR) documents the results obtained from a single execution of the Kubernetes Autoscaling Benchmark (KASBench) conducted from October 14, 2026 at 12:16:29 UTC to October 16, 2026 at 02:17:54 UTC. In this evaluation, the following autoscaler configurations were tested:

- Kubernetes Horizontal Pod Autoscaler (HPA)
- Kubernetes Vertical Pod Autoscaler (VPA)
- Kubernetes Event-Driven Autoscaler (KEDA)
- No autoscaler (control condition)

The purpose of this report is to completely document the benchmark environment, execution procedures, system configuration, and observed results of a full KASBench execution to enable independent verification and reproduction.

KASBench is a comparative benchmark intended to evaluate Kubernetes autoscaling systems under controlled workloads and documented conditions. It measures autoscaler performance using five metrics ($\theta^S$, $\theta^A$, $\tau^S$, $\tau^A$, and $\bar{C}$) and three composite scores ($D^S$, $D^A$, and $\zeta$). These metrics and composite scores measure deviation from the ideal, and are in the range 0 (best) to 100 (worst) inclusive. Lower scores represent autoscaler performance closer to the ideal.

**Synchronous underprovisioning penalty ($\theta^S$)** quantifies the mean deviation above a pre-determined service level objective (SLO) for synchronous requests initiated by a synthetic load generator over the benchmark period. Results are calculated as a percentage in the range 0 to 100, inclusive, using an upper bound of five-times the SLO. Sub-metrics for response-time ($\theta^S_{RT}$) and failure-rate ($\theta^S_{FR}$) are blended to calculate$\theta^S$.

**Synchronous underprovisioned time slices ($\tau^S$)** measures the percentage of 30-second time slices during which requests exceed the SLO by more than 20%. It is calculated by blending the sub-metrics for response-time ($\tau^S_{RT}$) and failure-rate ($\tau^S_{FR}$).

**Asynchronous underprovisioning penalty ($\theta^A$)** quantifies the how long asynchronous tasks wait in queue beyond a predetermined baseline (1 second). Asynchronous tasks are triggered indirectly from synchronous requests. Results are calculated as a percentage in the range 0 to 100, inclusive, using an upper bound of five-times the SLO.

**Asynchronous underprovisioned time slices ($\tau^A$)** measures the percentage of 30-second time slices during which asynchronous tasks exceed the baseline by more than 20%.

**Blended mean resource utilization ($\bar{C}$)** is the blended percentage of available CPU and memory requested by each autoscaler in excess of a baseline. The baseline is calculated as the minimum resources that could be allocated by any autoscaler in the test.

The metrics $\theta^S$, $\tau^S$, $\theta^A$, $\tau^A$, and $\bar{C}$ represent deviation from the ideal on a scale of 0 to 100, inclusive.

**Derived synchronous composite score ($D^S$)** is the weighted Minkowski deviation (distance) of $\theta^S$ and $\tau^S$ from zero.

**Derived asynchronous composite score ($D^A$)** is the weighted Minkowski deviation (distance) of $\theta^A$ and $\tau^A$ from zero.

**Composite score ($\zeta$)** is the weighted Minkowski distance of $D^S$, $D^A$, and $\bar{C}$.

The evaluation utilizes a completely random design. Each benchmark evaluation comprises multiple trials. The number of trials per autoscaler is indicated in the *Metadata* section of this document. Trials are independent. Each trial is conducted in an identical, individually built, one-time-use virtual private cloud in a randomly selected Amazon Web Services (AWS) availability zone. Autoscalers under test are randomly assigned to a trial and AWS availability zone.

## Autoscalers Under Test

This section describes the configuration of each autoscaler under test. Autoscalers are expected to be configured optimally for the benchmark evaluation, including resource requests and limits, number of replicas, and autoscaler-specific configuration.

### Kubernetes Horizontal Pod Autoscaler (HPA)

HPA is a threshold-based horizontal autoscaler. Each deployment is configured with a single replica. Initial CPU and memory requests are established by a calibration process using a fixed workload set at the $15^{\text{th}}$ percentile of benchmark workload. CPU and memory limits were set at 6 cores and 6 GiB, an arbitrarily large number to permit bursting. HPA was configured to allow scaling from 1 to 20 pods. CPU and memory thresholds are set at 70%. The scale down stabilization was set at 75 seconds.

### Kubernetes Event-Driven Autoscaler (KEDA)

KEDA is a threshold-based horizontal pod autoscaler driven by events or other metrics. In this evaluation, a hybrid configuration was employed, using KEDA for asynchronous microservices and HPA for others. KEDA monitored the depth of the Apache Kafka queues used for asynchronous processing. The lag threshold was set to 10 items in the queue. KEDA was configured with a minimum replica count of 1 and a maximum of 20 replicas. The polling interval was 15 seconds. The cool down period was set at 75 seconds. For asynchronous services, HPA was configured as described for the HPA-only evaluation.

### Kubernetes Vertical Pod Autoscaler (VPA)

VPA is a threshold-based vertical pod autoscaler. VPA initial resource requests and limits were the same as HPA. The minimum CPU was set the same as the requested CPU, as lower amounts were found to lead to instability. The minimum memory was set based on experimentation to avoid fatal out-of-memory errors. Each deployment was configured with two replicas to facilitate vertical scaling with no downtime. The Pod Disruption Budget was set at a minimum of 1 replica.

### No Autoscaler (control condition)

As the control condition, evaluations are conducted with no autoscaling configured. Deployments are configured with the same requests and limits as the other tests and a fixed replica count of 1.

## The Full Disclosure Report (FDR)

The following sections are produced systematically from data captured during the execution of the benchmark evaluation. All material parameters, settings, versions, hashes, and descriptions are included to foster reproducibility.

## Benchmark Metadata

| | |
|---|---|
| Benchmark Name | KASBench |
| Version | v1.0 |
| Testing Dates | 2026-10-14 12:16:29 UTC to 2026-10-16 02:17:54 UTC |
| start_time | 2026-10-14 12:16:29 |
| end_time | 2026-10-16 02:17:54 |
| Reporting Date | 2026-10-18 23:50:14 UTC |
| Region | us-east-1 |
| Benchmark Runner | c6a.large (amd64) |
| Control Plane | m8i.large (amd64) |
| Worker Nodes AMD | 1 x c6a.4xlarge |
| Worker Nodes ARM | 1 x c6g.4xlarge |
| AMD Worker AMI ID | ami-0244585558b2aebd7 |
| ARM Worker AMI ID | ami-052bbac83b5bb0ab9 |
| AMD Runner AMI ID | ami-03a891c9de365d954 |
| Kubernetes Version | 1.36.1 |
| Autoscalers Evaluated | Kubernetes Horizontal Pod Autoscaler (HPA), Kubernetes Event-Driven Autoscaler (KEDA), Kubernetes Vertical Pod Autoscaler (VPA), No autoscaler (control condition) |
| Trials Per Autoscaler | 7 |

# Software artifacts

## Benchmark Components

| Component | Image_original | ImageID |
|---|---|---|
| KASBench Controller | kasbench/kasbench-controller:0.0.6 | sha256:0816294976161de3279756584c5ee4061eafb860d80e436ca0a33dc99cedabc4 |
| KASBench Runner | kasbench/kasbench-runner:0.2.26 | sha256:8a130fc14b736fc3d6f2add2d305eef1e3a1703ade87cbb3f69efdf3796c1f6d |
| KASBench Load Generator | kasbench/kasbench-load-generator:0.0.21 | sha256:5e6ad9d312140a7f6e19f5792aa68b4a1c12d1e2c6999e98f0bd5f24549bca4d |

## Helm Charts

| Name | Version | URL |
|---|---|---|
| aws-ebs-csi-driver | 2.67.0 | https://kubernetes-sigs.github.io/aws-ebs-csi-driver |
| aws-efs-csi-driver | 4.5.1 | https://kubernetes-sigs.github.io/aws-efs-csi-driver/ |
| globeco | 0.1.60 | https://kasbench.github.io/globeco-helm |
| prometheus | 29.37.0 | https://prometheus-community.github.io/helm-charts |
| keda | 2.21.0 | https://kedacore.github.io/charts |

## Stateful Services

| Stateful Set | Image |
|---|---|
| postgres | postgres:18.0<br>docker.io/library/postgres sha256:41fc5342eefba6cc2ccda736aaf034bbbb7c3df0fdb81516eba1ba33f360162c |
| apache/kafka | apache/kafka:4.0.0<br>docker.io/apache/kafka sha256:3f7b939115cd4872e9cee9369d80bd69712fde55f9902f46d793f64848dedc75 |
| mongo | mongo:6.0<br>docker.io/library/mongo sha256:8b6d8f5bbedb25cb73517b65cf99f13aeb75ad5b157a56c479287a840bbad3ac |
| redis | redis:7-alpine<br>docker.io/library/redis sha256:858f009f9709ce576febc734aa78b8f6d624b82571f9ddb6bda4377c833b3499 |

## Stateless Services

| Deployment | Image |
|---|---|
| globeco-allocation-service | kasbench/globeco-allocation-service-server:1.0.3<br>docker.io/kasbench/globeco-allocation-service-server<br>sha256:0e6750db5d9e6227554a346f12d0556ea4c62a6047786f2395a539c74218cd5e |
| globeco-confirmation-service | kasbench/globeco-confirmation-service:1.0.11<br>docker.io/kasbench/globeco-confirmation-service<br>sha256:6275443ef40937002caba5156796c6a469b8665ac2bbae7d78b3009e646ffad7 |
| globeco-debug-tools | kasbench/globeco-debug-tools:latest<br>docker.io/kasbench/globeco-debug-tools<br>sha256:9f279ca1ccaf4d118c3de3587bb7946198ac6a14569e3990ea978fdc4f644032 |
| globeco-execution-service | kasbench/globeco-execution-service:1.0.9<br>docker.io/kasbench/globeco-execution-service<br>sha256:949ea19a52e2a118431f3241b9a10febdb154a406a499faa246194179abfd6a1 |
| globeco-fix-engine | kasbench/globeco-fix-engine:1.0.17<br>docker.io/kasbench/globeco-fix-engine<br>sha256:ae5cda2fdb69d92b94059cbead3677703c5f2a61da37659855b22d8d895c0450 |
| globeco-order-generation-service | kasbench/globeco-order-generation-service:1.0.6<br>docker.io/kasbench/globeco-order-generation-service<br>sha256:a7541d05c1e720573394cb41b443067a05bbb7db75e69e240df1808c5fed5f14 |
| globeco-order-service | kasbench/globeco-order-service:1.0.4<br>docker.io/kasbench/globeco-order-service<br>sha256:1600b3c42ace32ddf0296468173bc835344e4c885e0b528f4d4e71e51f457b7f |
| globeco-portfolio-accounting-service | kasbench/globeco-portfolio-accounting-service-server:1.0.2<br>docker.io/kasbench/globeco-portfolio-accounting-service-server<br>sha256:2b5646e0ccd9a5c4bf4bbce2749f6325c668802175cfc551c36c24fa36dee69d |
| globeco-portfolio-management-portal | kasbench/globeco-portfolio-management-portal:1.0.5<br>docker.io/kasbench/globeco-portfolio-management-portal<br>sha256:a65a6022cb244c4baf0eac2ad1f8dddf626ec4ecf0481947e639a0924feb1193 |
| globeco-portfolio-service | kasbench/globeco-portfolio-service:1.0.4<br>docker.io/kasbench/globeco-portfolio-service<br>sha256:b0561599453606420eb4e904f31214c9c62c5a580c5ca95b25e59f67eca16dd7 |
| | kasbench/globeco-portfolio-service-high-cpu:1.0.4<br>docker.io/kasbench/globeco-portfolio-service-high-cpu<br>sha256:1a7821e35baf1814f2a7fddbd446ca59b2deaa86fe0a822a5400a27fb59be49c |
| globeco-pricing-service | kasbench/globeco-pricing-service:1.0.2<br>docker.io/kasbench/globeco-pricing-service<br>sha256:2509c2a2e69712fa9a559b21a6f820ee537300592e6b77794c7135f282c06dc9 |
| globeco-security-service | kasbench/globeco-security-service:1.0.1<br>docker.io/kasbench/globeco-security-service<br>sha256:7066be0ea6f76bd60d1ba3c24a30e748c9c767a5eb13a2859204ab765e632241 |
| globeco-trade-service | kasbench/globeco-trade-service:1.0.6<br>docker.io/kasbench/globeco-trade-service<br>sha256:030719935159a3722ae29242134d4c4cff451ad2e82fcb85864d49aa604c9700 |

# Raw Performance Metrics

## Summarized Performance Metrics by Trial

| Trial ID | Autoscaler | Requests | Mean Response Time (ms) | Failures | Failure Rate | Round-trip Completion (%) |
|---|---|---|---|---|---|---|
| trial0001 | VPA | 538728 | 82 | 14 | 0.003% | 2.9% |
| trial0002 | HPA | 544340 | 76 | 11 | 0.002% | 1.2% |
| trial0003 | VPA | 549082 | 83 | 1006 | 0.183% | 2.5% |
| trial0004 | NONE | 437793 | 287 | 576 | 0.132% | 1.7% |
| trial0005 | HPA | 546222 | 76 | 12 | 0.002% | 1.2% |
| trial0006 | KEDA | 539007 | 80 | 112 | 0.021% | 6.6% |
| trial0007 | VPA | 538885 | 74 | 16 | 0.003% | 2.3% |
| trial0008 | VPA | 538097 | 79 | 7 | 0.001% | 3.0% |
| trial0009 | HPA | 542938 | 87 | 14 | 0.003% | 1.3% |
| trial0010 | NONE | 396881 | 237 | 223 | 0.056% | 2.8% |
| trial0011 | KEDA | 538791 | 84 | 133 | 0.025% | 6.2% |
| trial0012 | KEDA | 539247 | 92 | 105 | 0.019% | 7.0% |
| trial0013 | HPA | 541826 | 78 | 35 | 0.006% | 1.3% |
| trial0014 | KEDA | 540732 | 82 | 94 | 0.017% | 7.4% |
| trial0015 | KEDA | 531534 | 101 | 107 | 0.020% | 6.3% |
| trial0016 | NONE | 499510 | 279 | 447 | 0.089% | 1.3% |
| trial0017 | NONE | 400056 | 251 | 612 | 0.153% | 2.4% |
| trial0018 | KEDA | 543205 | 92 | 228 | 0.042% | 6.3% |
| trial0019 | HPA | 548058 | 83 | 75 | 0.014% | 1.4% |
| trial0020 | VPA | 547022 | 81 | 12 | 0.002% | 3.0% |
| trial0021 | KEDA | 532637 | 97 | 138 | 0.026% | 6.8% |
| trial0022 | NONE | 528991 | 166 | 59 | 0.011% | 1.2% |
| trial0023 | NONE | 524332 | 177 | 91 | 0.017% | 1.6% |
| trial0024 | HPA | 552611 | 73 | 19 | 0.003% | 1.1% |
| trial0025 | VPA | 548588 | 79 | 15 | 0.003% | 2.6% |
| trial0026 | HPA | 544396 | 75 | 51 | 0.009% | 1.0% |
| trial0027 | NONE | 468007 | 271 | 548 | 0.117% | 1.6% |
| trial0028 | VPA | 545748 | 80 | 9 | 0.002% | 2.2% |

## Summarized Performance Metrics by Autoscaler

| Autoscaler | Mean Response Time (ms) | Failure Rate | Round-trip Completion (%) |
|---|---|---|---|
| HPA | 78 | 0.006% | 1.2% |
| KEDA | 90 | 0.024% | 6.6% |
| NONE | 238 | 0.082% | 1.8% |
| VPA | 80 | 0.028% | 2.6% |

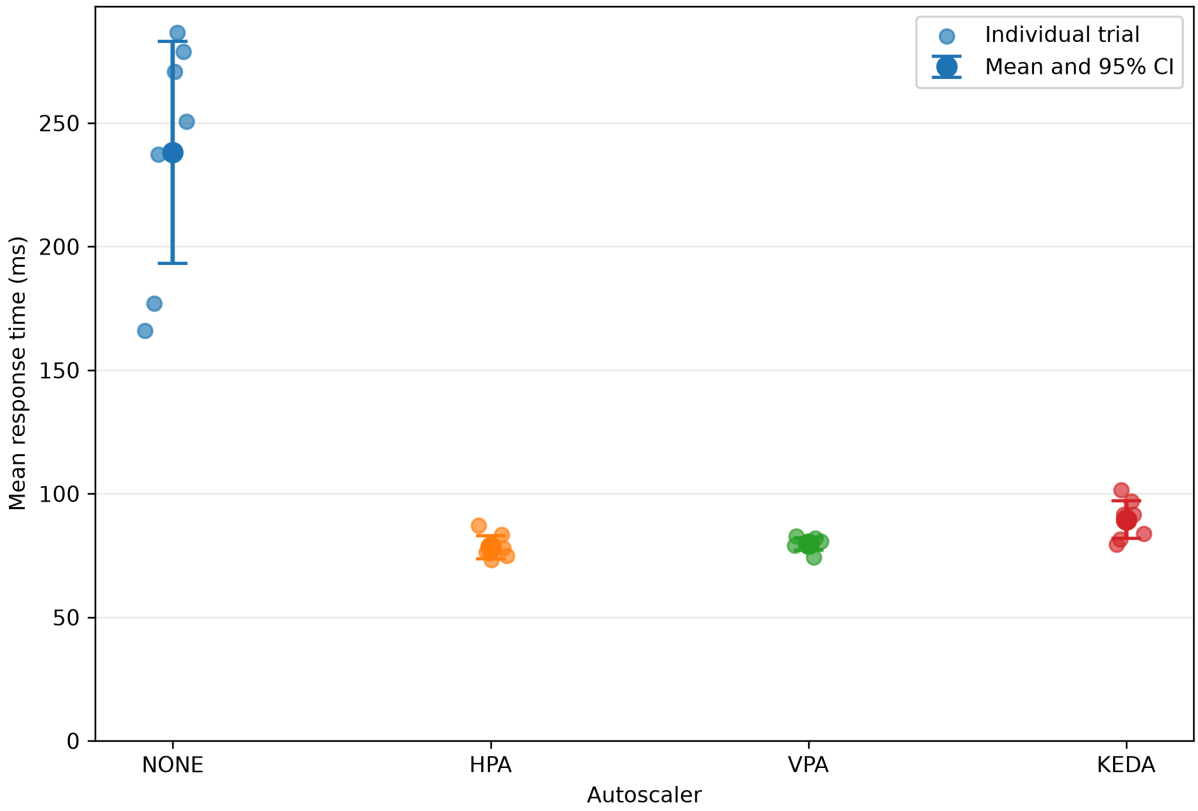



Figure 1. Mean response time by autoscaler.

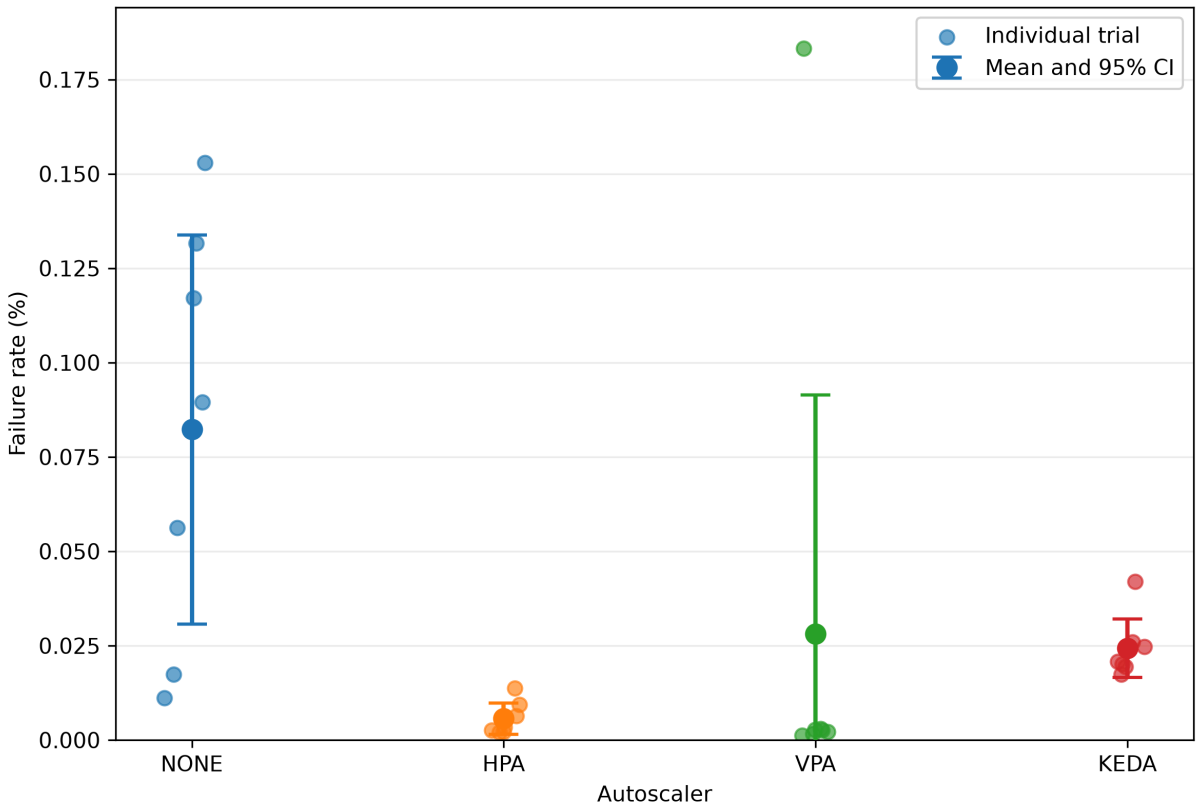



Figure 2. Mean failure rate by autoscaler.

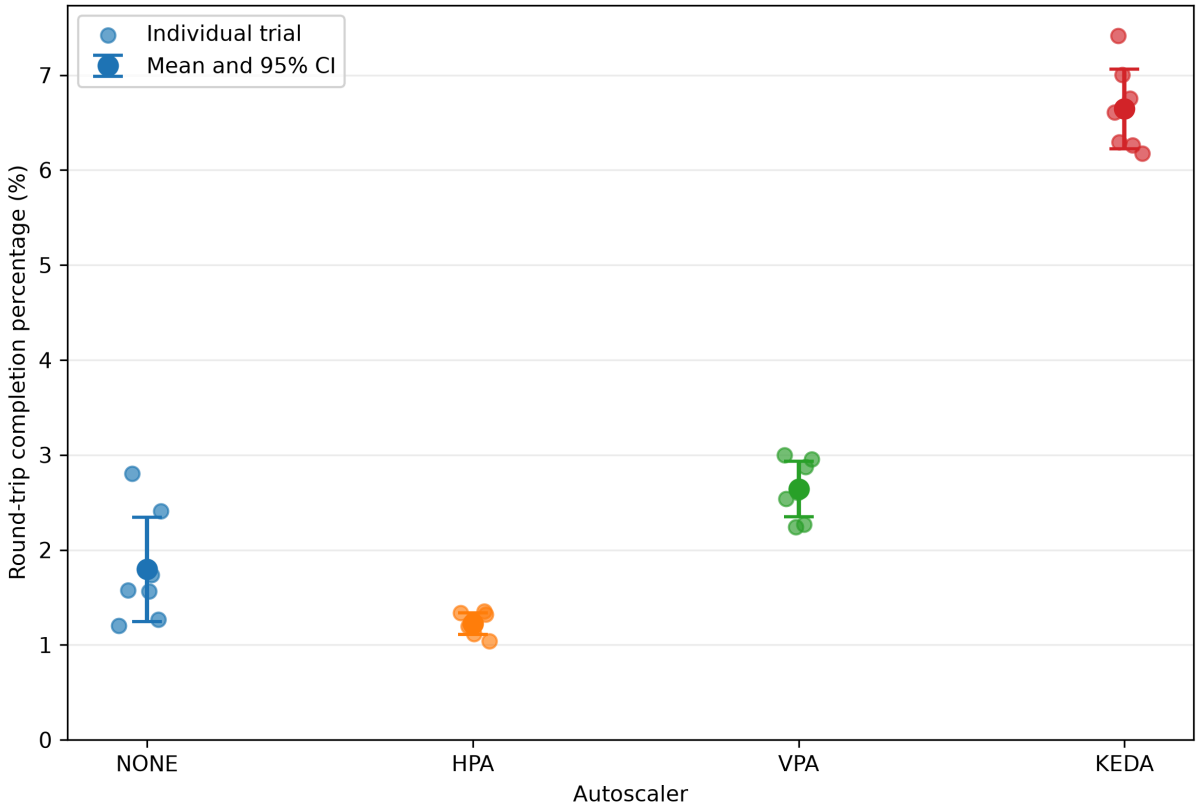



Figure 3. Mean round trip percentage by autoscaler.

## Comparative Autoscaler Metrics

### Synchronous Underprovisioning Penalty ($\theta^S$) by Autoscaler ($\alpha = 0.5$)

| Autoscaler | $\theta^{RT}$ | $\theta^{FR}$ | $\theta^S$ |
|---|---|---|---|
| HPA | 40.646 | 1.576 | 21.111 |
| KEDA | 42.913 | 3.204 | 23.059 |
| NONE | 71.924 | 15.165 | 43.544 |
| VPA | 45.554 | 1.434 | 23.494 |

## Synchronous Underprovisioned Timeshare ($\tau^S$) by Autoscaler ($\alpha$ = 0.5)

| Autoscaler | $\tau^{RT}$ | $\tau^{FR}$ | $\tau^S$ |
|---|---|---|---|
| HPA | 89.011 | 2.599 | 45.805 |
| KEDA | 88.430 | 4.132 | 46.281 |
| NONE | 91.371 | 19.846 | 55.608 |
| VPA | 89.138 | 2.715 | 45.927 |

## Asynchronous Metrics ($\theta^A$ and $\tau^A$) by Autoscaler

| Autoscaler | Avg. Lag | $\theta^A$ | $\tau^A$ |
|---|---|---|---|
| HPA | 13345.902 | 49.989 | 50.000 |
| KEDA | 50.379 | 12.821 | 15.486 |
| NONE | 7481.291 | 47.575 | 48.068 |
| VPA | 266.364 | 31.697 | 35.013 |

## Cost Metrics by Autoscaler ($\beta$ = 0.5)

| Autoscaler | $\overline{C}_{CPU}$ | $\overline{C}_{MEM}$ | $\overline{C}$ |
|---|---|---|---|
| HPA | 41.620 | 37.314 | 39.467 |
| KEDA | 55.485 | 45.451 | 50.468 |
| NONE | 13.653 | 14.614 | 14.134 |
| VPA | 42.135 | 36.088 | 39.112 |

## Summary Results by Autoscaler

| Autoscaler | $\theta^S$ | $\tau^S$ | $\theta^A$ | $\tau^A$ | $D^S$ | $D^A$ | $\overline{C}$ | $\zeta$ |
|---|---|---|---|---|---|---|---|---|
| HPA | 21.111 | 45.805 | 49.989 | 50.000 | 37.505 | 49.995 | 39.467 | 43.688 |
| KEDA | 23.059 | 46.281 | 12.821 | 15.486 | 38.189 | 14.278 | 50.468 | 36.631 |
| NONE | 43.544 | 55.608 | 47.575 | 48.068 | 50.300 | 47.823 | 14.134 | 45.664 |
| VPA | 23.494 | 45.927 | 31.697 | 35.013 | 38.011 | 33.437 | 39.112 | 36.565 |

## Parameters

| Parameter | Value |
|---|---|
| $w_{\theta A}$ | 0.5 |
| $w_{\tau A}$ | 0.5 |
| $w_{\theta S}$ | 0.5 |
| $w_{\tau S}$ | 0.5 |
| $w_S$ | 0.4 |
| $w_A$ | 0.4 |
| $w_C$ | 0.2 |
| p | 3 |

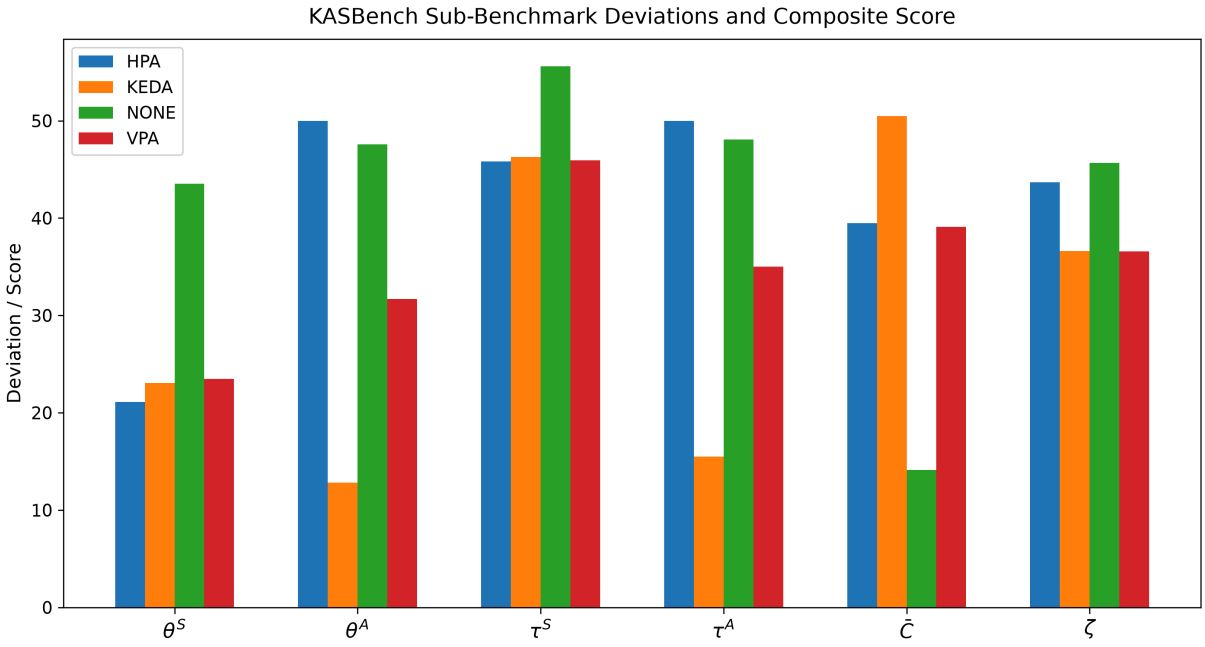



Figure 4. Sub-benchmark deviations and composite score by autoscaler.

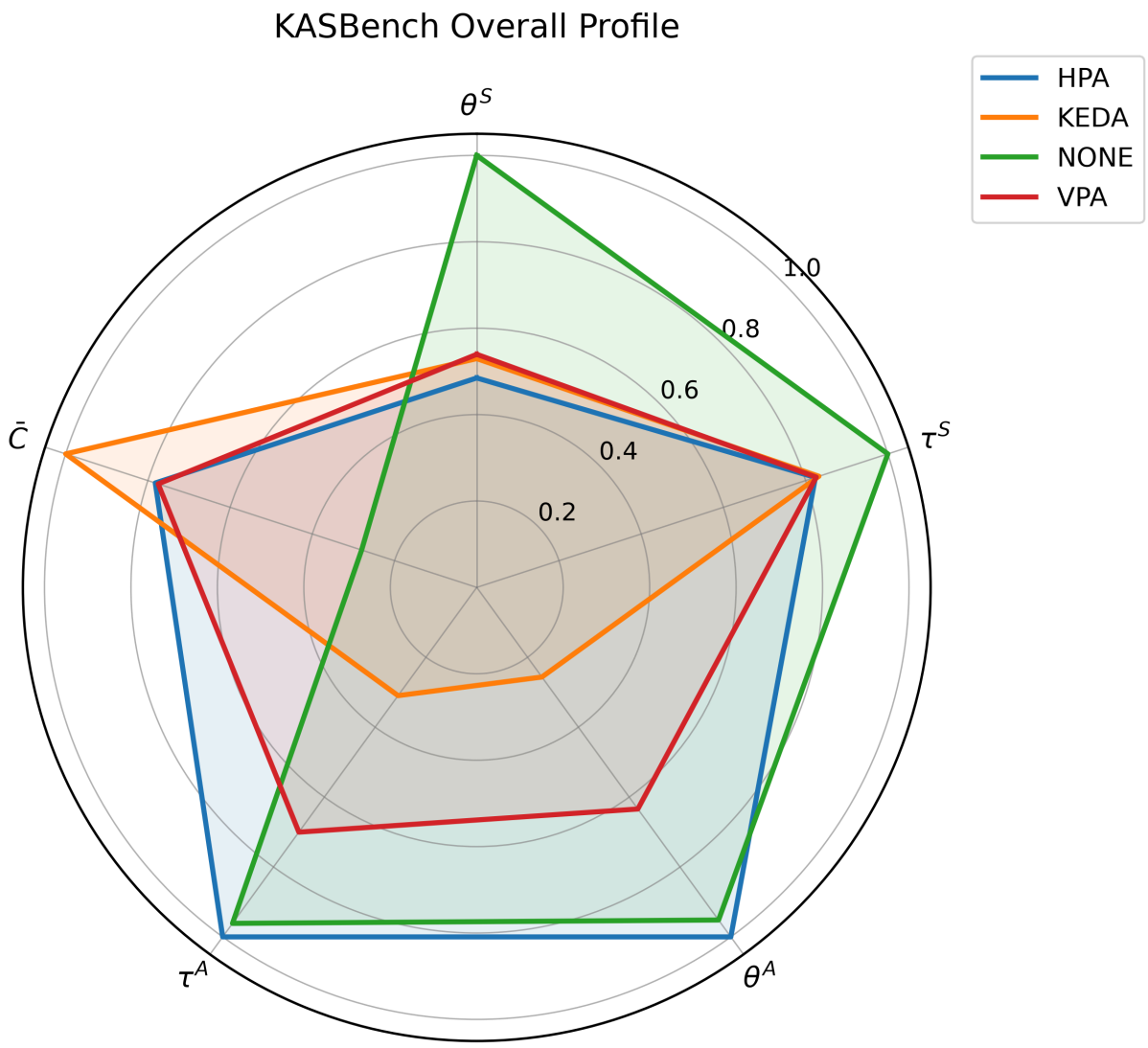



Figure 5. Sub-benchmark deviations by autoscaler.